9/10/2026

Renewal - reward Process

Let $\{G(i)\}$ be a series of i.i.d finite r.v.s. Then process

$$Y(t) = \sum_{n=1}^{N(t)} G(n)$$

is a renewal-reward process

- $Y(t)$ is defined by $\{A_i\}$ and $\{G_i\}$
- Pairs $\langle A(i), G(i) \rangle$ must be i.i.d, but $A(i)$ and $G(i)$ can be correlated
  - In particular, $G(i)$ can be a function of $A(i)$

Long-run Reward Rate

If $0 < \mathbb{E}[A_i] < \infty$ and $\mathbb{E}[|R|] < \infty$, then

$$\lim_{t\to\infty} \frac{Y(t)}{t} = \frac{\mathbb{E}[R_i]}{\mathbb{E}[A_i]} \quad \text{almost surely}$$

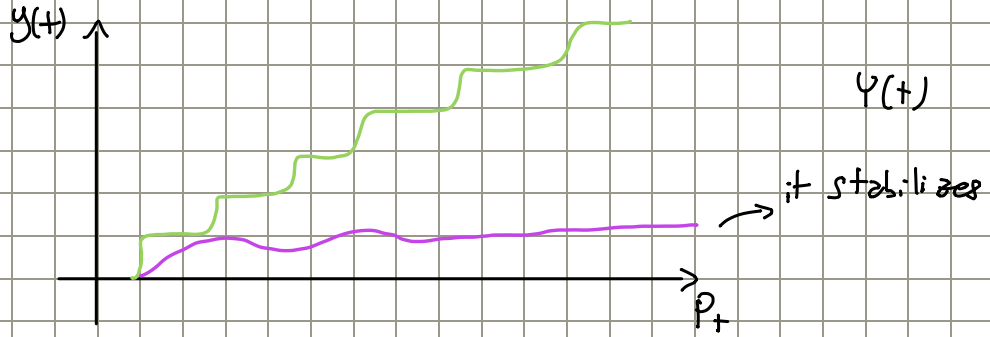
For traffic: $G_i = \sum_{j=1}^{B_i} W_{i,j}$. Thus long run offered rate is $\frac{\mathbb{E}[R_i]}{\mathbb{E}[A_i]}$

When batch size and individual weights satisfy the relevant independence assumptions can its numerator simplify to $\mathbb{E}[B_i]\,\mathbb{E}[W_{i,j}]$

$Y(t)$ = cumulative reward at time t

$\frac{Y(t)}{t}$ = average reward rate up to time t

$\lim_{t\to\infty} \frac{Y(t)}{t}$ = long-term average reward rate



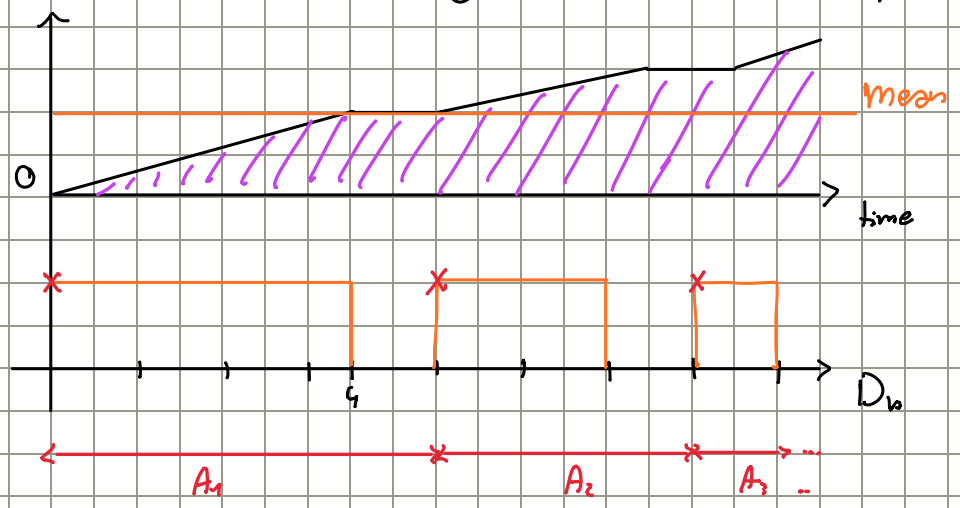
Example: Industrial machinery

Produces revenue of K €/day. Every x days, with $x \sim G(P)$ it needs to stop for maintenance, $\{x_n\}$ are i.i.d

→ It stops for y days, $y \sim U\{1,2,3\}$

We want to...

w_n = state of machine at day x = $\begin{cases} 0 & \text{if under maint} \\ 1 & \text{if operational} \end{cases}$



Is $\{w_n\}$ a Renewal Reward process?

$A_i = x_i + y_i$ Are iid?

↳ $G_i = K \cdot x_i$ gain after each interval

(A_i, G_i) is indep of (A_j, G_j) $j \neq i$

$$Y = \frac{E[G_i]}{E[A_i]} = \frac{E[K \cdot x_i]}{E[x_i + y_i]} = \frac{K \cdot E[x_i]}{E[x_i] + E[y_i]} = \frac{K \cdot \frac{1}{P}}{\frac{1}{P} + 2} * \frac{1+2+3}{3} =$$

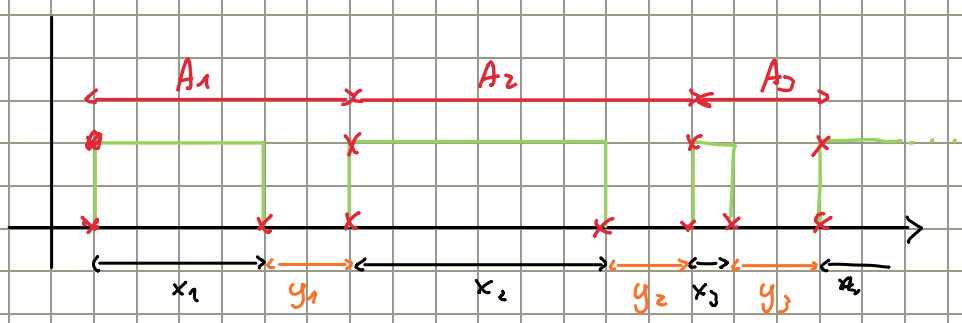
Now, same problem, with differences

Produces revenue of K €/day. Every x days, with $x \sim G(P)$ it needs to stop for maintenance, $\{x_n\}$ are iid

→ It stops for y days, $y \sim \begin{cases} 1 & \text{if } x_n = 1 \\ 2 & \text{if } 1 < x_n \leq 4 \\ 3 & \text{otherwise} \end{cases}$

During has a loss of h €/day

- Same model, we still have a process with active/lower state



$A_i = x_i + y_i$, A_{i+1} It's a renewal process

$G_i = k x_i - h y_i$ (A_i, G_i) Are independent

We can apply renewal-reward

$$\bar{Y} = \frac{E[G_i]}{E[A_i]} = \frac{E[kx_i - hy_i]}{E[x_i + y_i]} = \frac{E[kx_i] + E[-hy_i]}{E[x_i] + E[y_i]} = \frac{k \cdot \frac{1}{p} - h \cdot m_y}{\frac{1}{p} + m_y}$$

→ splittable because E is linear

$m_y = ?$ $E[x_i y_i] = E[x_i]\,E[y_i]$ iff $E[x_i]\,E[y_i]$

$$m_y = E[y] = \sum_{a \in \{1,2,3\}} a \cdot P[y = a] = \sum_{a=1}^{3} a \cdot P[x \in S_a]$$

$S_1 = \{1\}$; $S_2 = \{2,3,4\}$; $S_3 = \{5, \dots\}$

$$= p + 2p\left(1-p + (1-p)^2 + (1-p)^3\right) + 3(1-p)^4$$

$X \sim G(p)$

$P[x = k] = (1-p)^k p$; $k = 1, 2, \dots$

$P[x \in S_1] = p$, $P[x \in S_2] = P\left[(1+p)^2 + (1-p)^2 + (1-p)^3\right]$

$P[x \in S_3] = \sum_{k=5}^{\infty} (1-p)^q \cdot \sum_{j=0}^{\infty} (1-p)^d$ → $\frac{1}{p}$

$$\sum_{k=0}^{n} q^k = \frac{1 - q^{n+1}}{1-q}$$

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

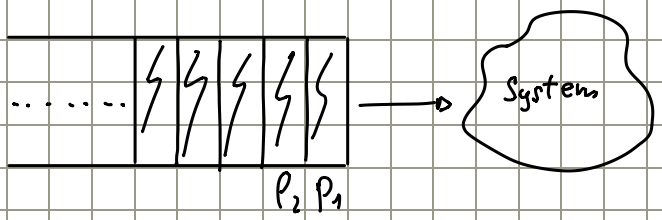
COMMON TRAFFIC MODELS

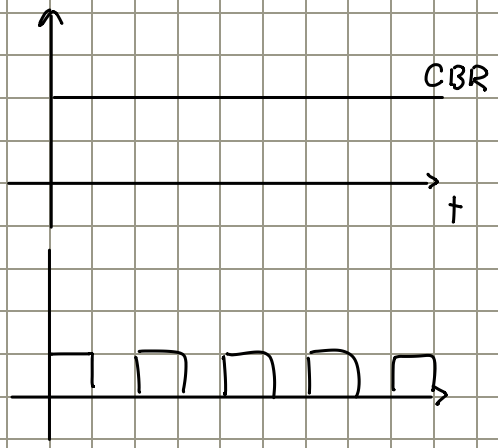
Saturated Source Model

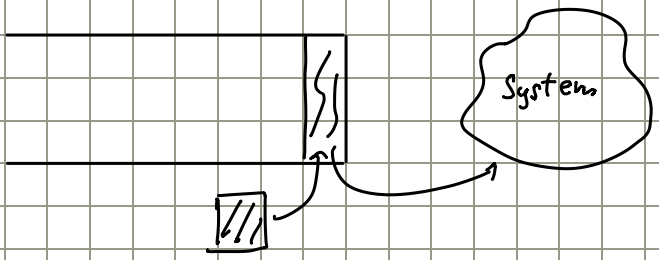
- source is always backlogged

- access limiting performance of a network in heavy traffic conditions
  - estimate max throughput capacity
  - no good for delays!

Saturated source







Sources that generate constant-rate data streams

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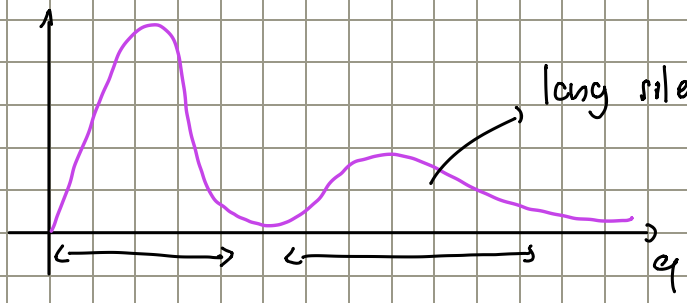
VBR source model

- Sources that generates traffic of variable rate over diverse time intervals
- Note: a Poisson is a common

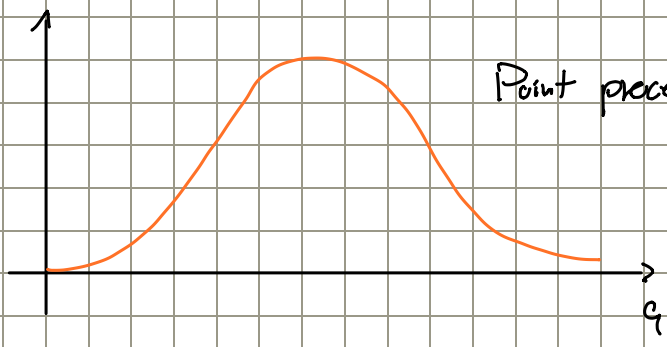
Bursty source models: what

Burst is a group of consecutive packets with interpacket gaps shorter than gaps to preceding and succeeding packets

- {A(i)} process is formed by runs of short values spaced by long time intervals
- Burstiness is a characterization of bursts in a flow



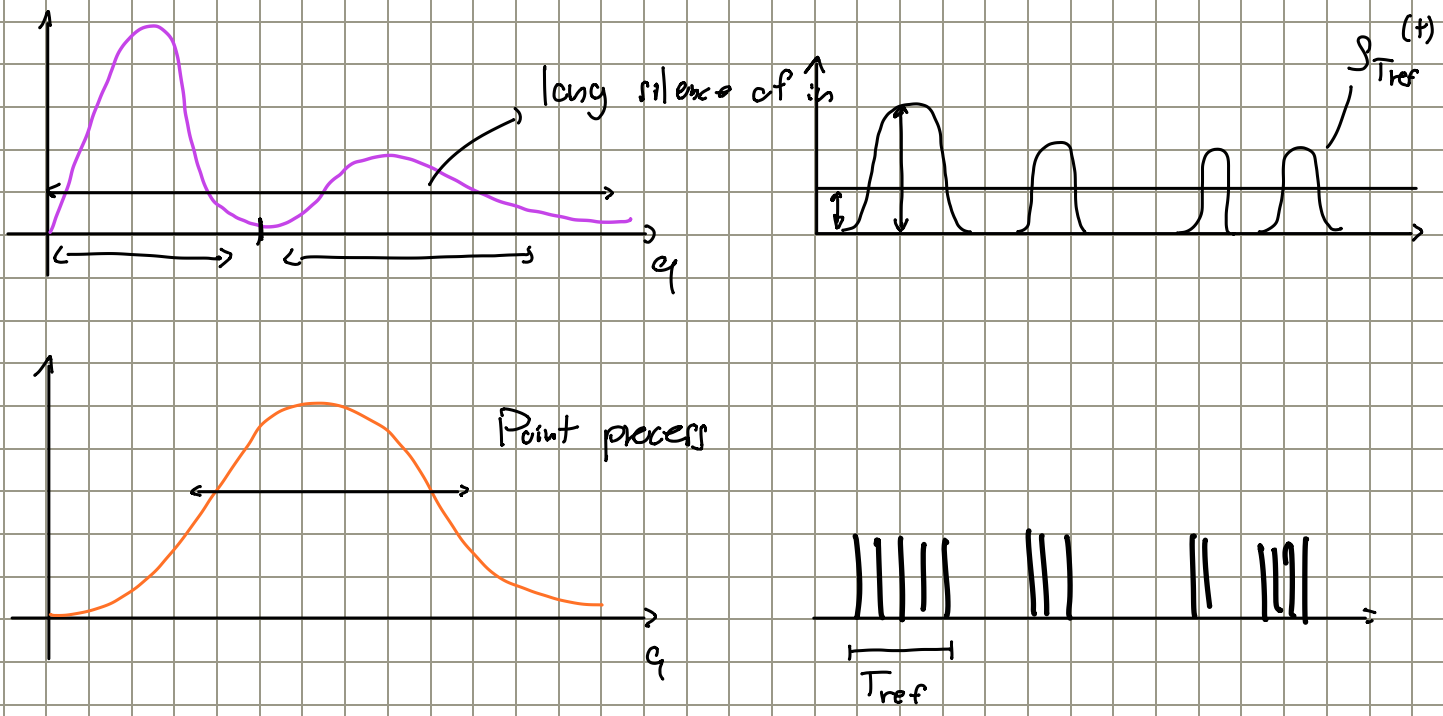






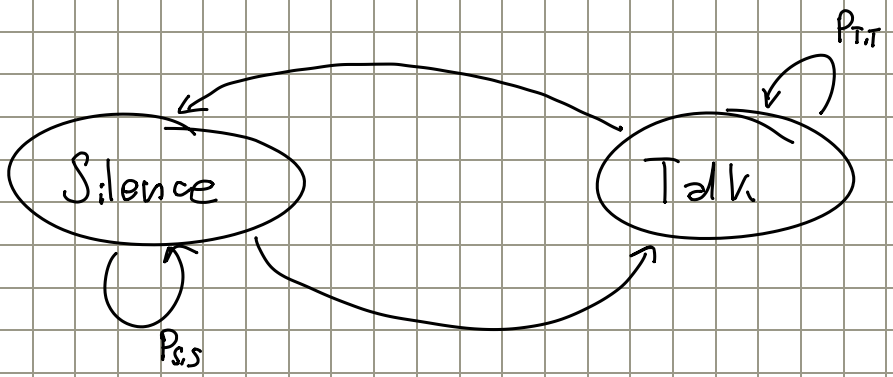
CBR or VBR sources that alternate periods

Formal definitions (also more intuitive one from professor)



Markov-Modulated Poisson Process (MMPP)

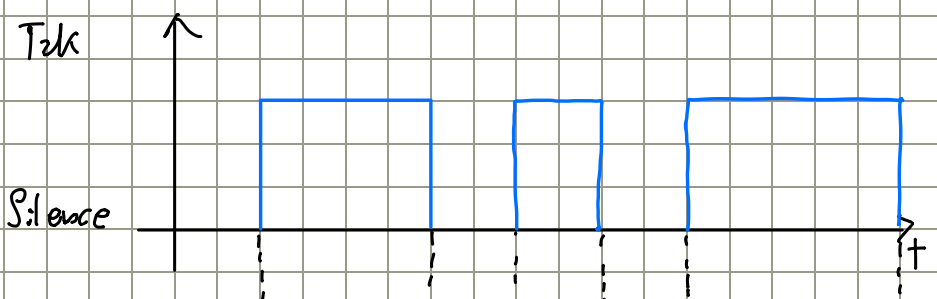
- Source can be in different "states" ($s \in S$)
- In each state s packets are generated according with a Poisson process of rate λ_s [pck/s]
- Workload can depend on state s
- States change in time according to an underlying Markovian Process

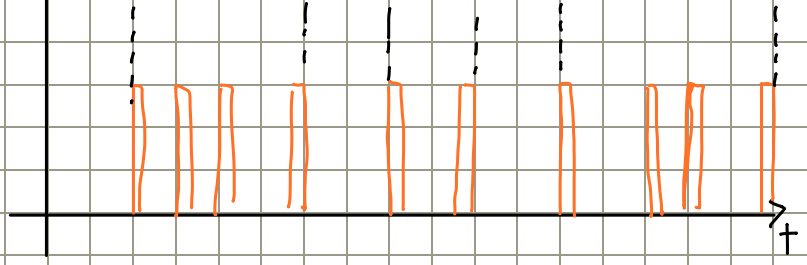


$x \in \{Silence, Talk\}$

if x = Silence
→ Poisson Process of rate $\lambda_S = 0$

x = Talk
Poiss. proc $\lambda_T > 0$





On - off Markovian model

OFF → no traffic

ON → CBR or VBR traffic

These models are very simplified

Homework 1 (check PDF)